Curriculum Vitae – Stefan Huber

Personal data

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Summary: Profile overview

Job Profile: DSP/SW/ML/AI **Speech+NLP Engineer**, Tech Lead, Data Scientist, PM/PO, all.
Objective: Pragmatically advance simple, efficient & robust data+model-centric AI solutions.

Working:

> 4 years	**Conversational AI, Voice Assistants**, PM + tech lead for many NLP/Speech use cases
2 years	**Speaker Verification & Anti-Spoofing** using Kaldi
1 year	Remote consulting as research engineer to **speech synthesis** companies
4 years	**DSP Audio Software** Engineer in the embedded area
1 year	Electronic Technician (PCB assembly line)

Education:

5 years	Ph.D. in **Voice Conversion** and Transformation (parametric speech synthesis, glottal source)
1 year	M.Sc. in **Sound and Music Computing** (time-stretching and pitch-shifting, phase vocoder)
4 years	B.Sc. of Honours in Computer Science (monophonic F0 estimation, **Speech 2 MIDI**)
3 years	Apprenticeship as **Electronic Technician** (analogue & digital circuits)

Professional experiences

2024 - Senior AI Specialist, Lean FS, Zürich, Switzerland (part-time)

AI agent framework to automate data adaptation/curation for LLM fine-tuning to optimize pruned models for various GenAI/STT/TTS uses cases for production.

2023 - 2024 Senior ML Engineer for NLP, AInstAIn, Zürich, Switzerland (own startup, failed)

Project design + prototyping of a deterministic + causal ML framework for trustworthy + transparent **xAI**: Reliable, robust, efficient, logic + symbolic at scale.

2020 - 2023 Voice / Speech Technologies Expert, AI & DS Team, UBS Zürich, Switzerland

- **Speech-as-a-Service** (NLT = **STT**, NLU + NLG, **TTS**): Design & streamline inbetween teams worldwide to buy vs build tech stacks for different jurisdictions and domain specific use cases with (a mix of) different languages-dialects-accents. Systematic & synthetic adaptation of **STT & NLU models** to cover the acoustic & language subspace of banking jargon as **data-centric AI customization** robustly. Customer satisfaction tool: Combine **emotion recognition** (audio) with **sentiment analysis** (text) to track client vs agent behaviour over time from calls.

- **Tech lead** of data scientists & junior engineers: Guidance in **NLP+Speech** tasks.

- **Stakeholder management**: Align tech concepts between senior management, business stakeholders and development teams to drive projects in various streams.

- **Service Owner** of **MS Azure Speech**: Cloud architectural (security) designs & risk assessments by service curation & maintenance for regulatory compliance audits. Design service for cloud red zone approval to run in production systems.

- **AI Model Governance** and vendor model validation for ethical + responsible AI.

2018 - 2019	Speech Research Engineer at SPitch, Zürich, Switzerland **Speaker Verification** (SV) using the SPitch solutions developed on top of **Kaldi.** - Proof-of-Concept **Anti-Spoofing deepfake detection tool** for replay attacks: Optimized 2 CNNs in PyTorch. Attention-based filtering with a U-Net architecture and ResNet classifier using dilated residual convolutions and squeeze-excitation networks. Measurable Performance: ~**92-96 % accuracy** on perfect lab conditions. - **PLDA optimization**: a) Scoring with combined means of VoicePrint and Score Averaging in context of multi-session enrolment and duration variability. Measurable Performance: **~20-30 % accuracy gain** given various data setups. b) Improved adaptation from rich Out-Of-Domain to low resource In-Domain corpora with a limited amount of data and speakers: **~8-15 % accuracy gain.**
2017	Speech Scientist Consultant for vocalID Inc., Boston, MA, USA Glottal excitation source estimation on different recording environments and speaker types; **Voice transformation** within acoustic space of one voice identity.
2016	Speech Scientist Consultant for ReadSpeaker AB, Uppsala, Sweden **Pitch transposition** using a phase-locked vocoder and spectral peak classification.
2015	Post-Doctorate at the Sound Analysis / Synthesis Team, Institut de Recherche et Coordination Acoustique / Musique (IRCAM, www.ircam.fr) in Paris, France Integration + adaptation of my PhD **speech analysis-transformation-synthesis** system into the singing voice synthesizer of the project 'ChaNTeR'.
2010 - 2011	Research Associate at IRCAM in Paris, France Project ircamAlign - **Automatic phonetic labelling** and phoneme border detection: Porting of HMM-based sequence alignment algorithm from Linux to Mac OS X.
2009	Internship at Yamaha Music Instruments in Hamamatsu, Japan **Singing voice synthesizer** www.vocaloid.com: High-level phoneme duration model to improve the synthesis quality of fast singing sequences defined by users.
2006 - 2008	DSP Audio Software Engineer at Teleca Systems in Nuremberg, Germany Implementation of logical driver SW for audio and GSM Layer1 communication.
2004 - 2006	DSP Audio Software Engineer at Philips in Leuven, Belgium Porting of **Speech Enhancement** DSP modules: Echo Cancellation + Noise Suppression; Signal-Noise-Ratio estimation; Comfort Noise Injection; Spectral Enhancement. Responsible for releases on TI DSP platforms to customer Samsung.
2004	Audio Software Developer at ALCATech in Dresden, Germany Implementation of an Analog-Digital converter module in Borland Delphi.
2001 - 2002	Internship at the R&D Centre of Texas Instruments in Bangalore, India Development of a **split-radix FFT**, MDCT and IMDCT function. Porting from floating- to fixed-point arithmetic. Integration on a TI DSP platform in C / ASM.
1997 - 1998	Community Service at a home for handicapped people in Polsingen, Germany
1996	Industrial Electronic Technician at Alcatel SEL in Gunzenhausen, Germany

Academic and educational background

2010 - 2015

Ph.D. at IRCAM and Université Pierre et Marie Curie, Sorbonne, Paris 6, France

Thesis topic: "Voice Conversion by modelling and transformation of extended voice characteristics"
3 major contributions:
1. Estimation and transformation of the glottal excitation source
2. A parametric speech analysis-transformation-synthesis system
3. A concatenative synthesis based VC system using whole phoneme units

2008 - 2009

M.Sc. in Sound and Music Computing, Music Technology Group,
Universitat Pompeu Fabra, Barcelona, Spain

Grade in percentage: 85.42 %
Grade Point Average (GPA): 3.41667
Thesis topic: "Harmonic Audio Object Processing in Frequency Domain"
(Time-Stretching and Pitch-Shifting employing Phase Vocoder techniques in a MIDI-supervised single-channel polyphonic audio stream)

Classes: Advanced Topics in Artificial Intelligence, Advanced Topics in Sound and Music Computing (Journal Club), Audio & Music Analysis, Audio & Music Processing, Computational Neuroscience, Music Perception & Cognition.

1998 - 2003

Bachelor of Honours in Computer Science at FHTW in Berlin, Germany

Grade in percentage: 84.5 %
Grade Point Average (GPA): 3.12
Thesis topic: Monophonic F0 estimation ('convert' Voice to MIDI notes in C++)

Subject Digital Audio Signal Processing: Implementation of different filters like lowpass, highpass, bandpass, bandstop, shelving, peak and notch filters.

1996 - 1997

Technical College in Weißenburg, Germany (A-Level)

1992 - 1996

Apprenticeship as Electronic Technician at Alcatel in Gunzenhausen, Germany

Analogue and digital electronics; bipolar, combinatorial and integrated circuits; RLC networks; transistors; operational amplifiers; AD-DA converters.

Publications

- Conference paper, Interspeech ISCA 2015:
"On glottal source shape parameter transformation using a novel deterministic and stochastic speech analysis and synthesis system"

- Conference paper, Sound and Music Computing Conference (SMC) 2015:
"Voice quality transformation using an extended source-filter speech model"

- Journal article, Computer, Speech & Language 2013:
"On the use of voice descriptors for glottal source shape parameter estimation"

- Conference paper, Interspeech ISCA 2012:
"Glottal source shape parameter estimation using phase minimization variants"

- Conference paper (Co-Author), ICASSP 2012:
"Analysis and modification of excitation source characteristics for singing voice synthesis"

IT Skills

Programming:	Language:	Skill level: 0=none, 10=perfect
	Python	9
	Bash	8
	Matlab	7
	Perl	6
	C/C++	5

AI frameworks: PyTorch, TensorFlow, LangChain-Graph-Smith, Gradio, Vector DBs, OpenAI API
Cloud tooling: MS Azure, Amazon AWS+EC2, GCP / Docker+Kubernetes, mlflow, WnB, REST
Editors, Compilers: Komodo Edit, Sublime Text, Emacs, Matlab, Eclipse, MS Visual Studio
Version systems: BitBucket, CI/CD/CT/CM, CVS, Git, Gitlab, SVN
PM processes: Agile, Kanban, Lean, Scrum, Waterfall

MOOC class 2025: HF Agents: Fundamentals
MOOC class 2024: **Langchain:** **GenAI** Text Summarization, Chat with Data / Agents: **AI Agents**, Build DB Agent / LLMs: Fine-Tuning LLMs, **LLMOps**, Fine-Tune BERT for Text Classification, **Semantic Search**, Transfer Learning for NLP / **RAG**: Agentic RAG with LlamaIndex, Knowledge Graphs for RAG / **Vector DBs**: From Embeddings to Apps, Build Apps / Quantization: Fundamentals, In-Depth / Build an AutoEncoder / GenAI: Apps with Gradio, Debugging with Weights+Biases / OS models with HF
NVIDIA DLI 2023: **Data + Model Parallelism:** **Train on Multiple GPUs** + Build+Deploy Large NNs / Build Conversational AI Apps, Build Transformer-Based NLP Apps
IBM Workshops: Neuro-Symbolic AI Essentials 2023, Neuro-Symbolic AI Summer School 2024
MOOC class 2023: Microsoft AI-900 **Azure AI** Fundamentals
MOOC class 2023: Microsoft AI-102 **Azure AI** Engineer Associate
MOOC class 2023: **MLOps** 3: ML Modeling Pipelines in Production
MOOC class 2022: **MLOps** 2: ML Data Lifecycle in Production
MOOC class 2022: **MLOps** 1: Intro to ML in Production
MOOC class 2022: **Agile Project Management**
MOOC class 2022: Bitcoin and Cryptocurrency Technologies
MOOC class 2022: Build Basic Generative Adversarial Networks (**GANs**)
MOOC class 2021: Microsoft AZ-900 **Azure** Fundamentals
MOOC class 2021: Natural Language Processing with Attention Models
MOOC class 2019: DeepLearning.ai
1 NNs+DL / 2 Hyperparameter Tuning, Regularization, Optimization / 3 Structure ML Projects / 4 CNNs / 5 Sequence Models
MOOC class 2018: Neural Networks for Machine Learning, University of Toronto
MOOC class 2016: Machine Learning, Stanford University

Languages

German: Native
English: Fluent
Spanish: Good
French: Good
Dutch: Intermediate

Hobbies, social skills, social media

Hobbies: Jogging, Swimming, Hiking, Skiing, Diving; Lindy Hop dancing; Conscious cooking

Soft skills: Collaborative co-worker / Resilient, resistent to stress / Self-reflexive, open to constructive feedbacks, self-confident, emotional intelligence / Intuition, imagination, innovation / practically-oriented results-driven problem solver, get hands dirty in stretch assignments

Social + coding: www.linkedin.com/in/huberstefan, github.com/StefanGitHuber, http://bit.ly/4iGqJhc

Further references, certificates and diploma open request